Concept Drift Detection on Unlabeled Data Streams: A Systematic Literature Review

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***Abstract*—Dynamic data streams applications are bound to potential changes in data distribution, of which in the context of data stream mining, will cause concept drift. Data stream mining model must have the capability to adapt to concept drift, otherwise, risk the deterioration of its learning performance. Most of the existing surveys on concept drift detection methods focused on labeled data streams, that may be inapplicable to scenario where true labels are unavailable. The aim of this systematic literature review is to study the existing concept drift detection methods on unlabeled data streams, focusing on the learning process and the way concept drift is monitored in the data stream mining model. A total of 15 articles are selected for final analysis, and it is found that most of the drift detection methods were applied in a supervised learning setting. An experimental evaluation of the methods can be performed in the future work to investigate their performance in unsupervised learning setting.**

Keywords—data streams, concept drift, unlabeled, unsupervised

I. Introduction

The emergence of smart phones, Internet-of-things devices and sensor networks has contributed to an overwhelming and ever-increasing volume of data currently available in streaming format. The data instances in data streams applications arrive in sequence at a time-varying interval. Data stream mining involves the process of storing, processing, and extracting feasible knowledge from real-time data that can assist in decision making and understanding phenomenal events. A majority of standard data mining model are inapplicable to data stream mining due to three main features of data streams: (1) Size: The data instances arrive in bulk and can be infinite that it may be impossible for data stream applications to have sufficient available memory to store the data instances in their original forms. In such case, the data instances must be converted to a compact form using appropriate reduction method. (2) Speed: The high rate of arrival of data streams requires the data streams to be processed as soon as it arrives in a single scan. (3) Data distribution: Data streams are prone to various kind of changes in data distribution that should be dealt efficiently in case of its occurrence. Applications with potential changes in data distribution are prevalent. One real-world example involves a sensor networks that gathers data about physical events. The sensing units typically undergo irreversible ageing or defects in their built-in senses. The physical events that the sensors are monitoring can also change over time, resulting in different sensor readings. In these conditions, change in the data distribution is inevitable.

In data stream mining, the dynamically changing environments where the distribution of data streams can change over time cause a phenomenon called concept drift. The problem of concept drift was first proposed in [1] where the authors modelled a supervised learning task that drifted due to changing environment. Given a time period [0, t], a set of data streams, denoted as $S_{0,t} = \{d_0, \ldots, d_t\}$, where $d_i = (X_i, y_i)$ is a data element, X_i is the feature vector, y_i is the label, and $S_{0,t}$ follows a certain distribution $F_{0,t}(X, y)$. Concept drift occurs at timestamp $t + 1$ if $F_{0,t}(X, y) \neq F_{t+1,\infty}(X, y)$, denoted as $\exists t: P_t(X, y) \neq P_{t+1}(X, y)$. Accordingly, concept drift at time t can be defined as the change of the set of X that define y at time t. In order to avoid performance degradation of its learning algorithm, the data stream mining model must be able to handle and adapt to concept drift. The availability of data labels will affect the choice of method to handle concept drift [2]. If the true labels can be presented to the learning algorithm before the next data element arrives, performance-based indicators such as accuracy, recall, precision, sensitivity, and specificity can be used to detect concept drift. These indicators benefit from being accurate and independent of the learning algorithm. However, they require the availability of a true label which may in turn delay the detection of concept drift if the true label is not immediately available. When the true labels arrive with delay or only a small number of true labels is available, concept drift can be detected by monitoring the predicted label with suitable statistical indicators. With the absence of labelled data, distribution of the data instances is typically used to detect concept drift. In real-world data stream problem, it is often the case that the true labels are not immediately available or nearly unavailable for concept drift detection.

Concept drift is a widely studied issue in data stream mining research in the last decade with comprehensive surveys on concept drift detection methods can be found in [3 – 8]. With the exception of [8], most of the concept drift detection surveys focus on methods that assume true labels is immediately available or arrive with delay, very few research has been conducted to address unlabeled concept drift detection methods. A general criteria for designing concept drift handling approach is proposed in [7] that outlined five common questions: "*How are data processed in the application?*", "*How is the learning task processed?*", "*How is concept drift monitored*", "*How is concept drift handled?*", and "*What are the performance criteria to be considered?*". The survey in [8] proposed concept drift detection taxonomy based on the way data are processed within the application. In this paper, literatures on unlabeled concept drift detection methods are systematically reviewed with the intent of

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studying the learning process and the way concept drift is monitored in the data stream mining model of the existing literatures. This paper answered the following research questions: (1) How many studies are there that discuss on concept drift detection method using unlabeled data streams? (2) How can the methods be categorized in terms of their learning task and concept drift monitoring? (3) How does the concept drift detection methods work?

The remainder of this paper is organized as follows: The method used in this study is first explained followed by the analysis of the results and discussion on the findings are then elaborated. The last section concludes the paper.

II. Material and Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses, PRISMA which outlines four stages of review was adopted in this study. The related articles were gathered from selected databases and the articles that did not meet the criteria were excluded. The articles included for review were further examined by analyzing the titles and abstracts, and final articles that meet the eligibility criteria were selected for analysis.

A. Resources

The review was performed using five key databases: Scopus, Web of Science, IEEE, ACM and ScienceDirect. A total of 2100 articles pertaining to concept drift are indexed by Scopus, while a total of 1244 articles are indexed by Web of Science. The other three databases (IEEE, ACM, and ScienceDirect) indexes a total of 590, 116, and 262 articles respectively.

B. Identification

The important step in a systematic literature review is to construct relevant search string related to the topic of interest. Three key terms that are the common keywords in similar researches were used in this study. Based on the search strings listed in Table I, a total of 479 articles containing the key terms in its title, abstract, or keyword, were retrieved from all databases (Scopus: 203, Web of Science: 131, IEEE: 62, ACM: 53, ScienceDirect: 30). In addition, the study also performed manual searching and found 6 related articles, and 3 articles from [8] were also retrieved yielded altogether 488 articles for screening.

TABLE I. Search String

Database	Search String
Scopus	TITLE-ABS-KEY(("concept drift") AND ("unsupervised" OR "unlabeled"))
Web of Science	TOPIC: (("concept drift") AND ("unsupervised" OR "unlabeled"))
IEEE	("All Metadata":"concept drift") AND ("All Metadata":"unsupervised" OR "All Metadata":"unlabeled")
ACM	[Abstract: "concept drift"] AND [[Abstract: "unsupervised"] OR [Abstract: "unlabeled"]]
ScienceDirect	Title, abstract, keywords: ("concept drift") AND ("unsupervised" OR "unlabeled")

C. Screening

After the retrieval of articles from the selected databases, duplicate articles were removed in the screening phase. In total, 246 articles are identical and subsequently removed. The remaining 242 articles were assessed against the inclusion and exclusion criteria specified in Table II. This study focused on peer-reviewed journals and conference proceedings, excluding systematic review articles, surveys, workshop papers, and technical reports from the review. Furthermore, the study concentrated solely on articles written in English. In terms of the study timeline, the study expanded to a 16-year period (2004 – 2020) due to the frequency of articles published within the duration and to obtain detail information about the topic of interest. Overall, 14 articles that did not meet the inclusion criteria were omitted from the review.

TABLE II. Screening Criteria

Criterion	Inclusion	Exclusion
Source type	Journal articles, conference proceedings	Reviews, workshop papers, technical reports
Source language	English	Non-English
Timeline	2004 - 2020	<2004
Subject area	Computing	Non-computing

D. Eligibility

A total of 228 that meet the inclusion criteria were screened for eligibility. At this point, the titles and abstracts of the papers were examined in detail to ensure they meet the inclusion criteria and are suitable for use in the current review. Consequently, 187 articles were excluded as they concentrate on concept drift detection from labeled data streams and briefly tap on unlabeled concept drift. As a result, a total of 15 articles were selected for full-text reading and analysis.

III. Results

Preliminary findings showed that 41 articles were related to the detection of concept drift from unlabeled data streams. However, in-depth examination of the main contents further excluded a total of 26 articles. Most of the excluded articles were found to require partially labeled data in order to detect concept drift, while some of the articles did not consist of the learning process applications to test their concept drift detection method. To answer the first research question, this study found a total of 15 articles that can be analyzed in terms of their learning process and concept drift monitoring method. Table III detailed the categorization of the final articles by learning type, learning algorithms and drift indicator methods ordered by the most recent publications.

TABLE III. Analysis of Existing Literatures on Concept Drift Detection from Unlabeled Data Streams

Articles	Learning Type	Learning Algorithms	Drift Detection Method
[9]	Supervised	Support Vector Machine	Kolmogorov–Smirnov Test
[10]	Supervised	Adapting Hoeffding Tree	Kolmogorov–Smirnov Test
[11]	Supervised	Binary Classification	Jaccard distance
[12]	Supervised	Kernel Ridge Regression	Clustering Similarities
[13]	Supervised	Naïve Bayes, k-Nearest Neighbor, Support Vector Machine	Kolmogorov–Smirnov Test
[14]	Supervised	Binary Relevance, Label Powerset, Pruned Sets	Novelty Detection Test

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Articles	Learning Type	Learning Algorithms	Drift Detection Method
[15]	Supervised	Statistical Learning Theory	Plover Algorithm
[16]	Supervised	Active learning	Margin Density
[17]	Unsupervised	Gaussian-based Clustering	Chernoff Bound
[18]	Supervised	Weighted Extreme Learning Machine	Clustering Similarities
[19]	Supervised	Naïve Bayes, Hoeffding Tree, k-Nearest Neighbor	Confidence Vote
[20]	Supervised	Support Vector Machine, Linear Discriminant Analysis	Clustering Similarities
[21]	Supervised	k-Nearest Neighbor, Naïve Bayes, Decision Tree	Kolmogorov–Smirnov Test
[22]	Supervised	Support Vector Machine	Margin Density
[23]	Supervised	k-Nearest Neighbor	Confidence Vote

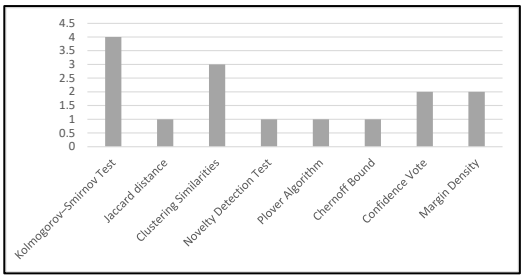


Fig. 1. Drift detection method by frequency of use.

Based on Table III, most concept drift detection on unlabeled data streams were applied in supervised learning setting with only one application on unsupervised learning. In general, concept drift was being detected from variety of classification algorithms based on Kolmogorov–Smirnov test, Jaccard distance, clustering similarities, novelty detection test, Plover algorithm, Chernoff bound, confidence vote, and margin density. In the order of frequency of use as visualized in Fig. 1, the Kolmogorov–Smirnov test, clustering similarities, confidence vote, and margin density are the top drift detection methods used in the literatures.

IV. Discussion

This section explained each of the concept drift detection methods listed in previous section:

A. Kolmogorov–Smirnov test

Kolmogorov–Smirnov test is a hypothesis test used to check if two samples are of the same distribution. The test depends on two metrics, *p*-value and significance value of the two samples. The *p*-value is a probability of an instance occurring. The significance value or alpha, α is a threshold value that determines whether the *p*-value will be rejected, or otherwise. Its value depends on the problem at hand, but the value is typically set to 0.05 or 0.01.

In concept drift detection, data streams arriving at a specific time are divided into data blocks of equal number of instances. The *p*-value for each two data blocks is calculated and compared with user-defined significance value. A *p*-value that is smaller than the significance value indicates that the two data blocks has different distribution, thus, signaling that concept drift occurs.

The test is sensitive to change, making it an effective method to detect concept drift event between two data blocks. It is model-independent and can be used on any learning algorithms without the need of data labels. However, it is designed to analyze one-dimensional data and cannot be easily extended for multi-dimensional data [24]. In real scenario, data streams are not limited to univariate data but may also arrive in multivariate streams.

B. Clustering Similarities

Clustering is an unsupervised machine learning method that groups instances of similar characteristics together. It is used in the pre-processing stage of supervised learning task to label unlabeled data streams. Standard *k*-means clustering was applied in [12] and [20] to obtain an initial clustering structure for unlabeled data streams. *k*-means clustering typically group instances into user-defined instances, *k*. Instances closer to any cluster centroid are grouped in the same cluster. Clustering similarities is utilized in concept drift detection by calculating the difference between subsequent unlabeled instances and the cluster centroids based on Euclidean distance. If the subsequent unlabeled instances deviate from the initial cluster set, a concept drift is signaled to the base supervised learning algorithm.

The fundamental issue that needs to be considered in standard *k*-means clustering is the definition of the optimal number of clusters. Poor definition of the number of clusters leads to inaccurate clustering of unlabeled data streams. *k*-means is sensitive to outliers that can be mistaken as concept drift. A challenge in data streams clustering is to distinguish whether the change in the data streams is due to concept drift or outlier. As the current concept drift detection from unlabeled data streams is limited to *k*-means clustering, it is crucial to study other type of clustering method that is capable of handling outliers.

C. Confidence Vote

Ensemble classifier is a supervised machine learning method that divide a dataset into several samples and independently create a base classifier for each sample. Each base classifier outputs a confidence vote that is used to generate a new classifier with a higher confidence vote than the base classifiers. To detect concept drift, the data streams is divided into several data blocks and the confidence vote of the base classifier for each data block is calculated. Significant change in the confidence vote signals a concept drift.

The concept drift detection based on confidence vote of ensemble classifier avoid the needs for labeled data to calculate prediction error. It is model independent and can be applied using any learning algorithms. Concept drift can be detected with smaller false alarm rate and detection misses. However, in the existing studies [19, 23], this method was only applied to categorical and one-dimensional data stream. Further exploration using this method can look into the effects of different types of data streams on the algorithm performance.

D. Margin Density

In a supervised learning problem, a decision boundary is the line on hyperplane that differentiate between classes. A margin is defined as the region near to the decision boundary

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containing instances with uncertain class. The margin density method relies on the classifier's margin to detect concept drift. It calculates the proportion of data instances in the margin and triggers a drift alert when the margin density exceeds density threshold.

This method is model-independent, and any learning algorithms can be applied as the base classifier. However, this method depends on the definition of density threshold that may varies in different applications and requires expert judgement. Moreover, the problem if differentiating between a concept drift and outliers in the margin is an important issue that needs to be tackled. Future work can explore on the evaluation of the margin density in detecting drifts from multi-classification problems [16] and the use of ensemble classifiers as the base classifiers to improve the algorithm accuracy [22].

E. Jaccard Distance

The Jaccard distance is a statistical measure of dissimilarity between data samples. To detect concept drift, the data streams is divided into a set of data blocks and the distance between instances in the data block is computed based on Jaccard similarity. A Jaccard similarity greater than the threshold value indicates concept drift. Instances with concept drift are removed, and the learning model is updated accordingly.

This method depends on a user-defined threshold that may affects the algorithm capability to detect concept drift and increase false alarm rates. In addition, the proposed method in [11] was applied to binary classifiers, thus having the constraint of imbalanced class in multi-class classification and window size which can be further explored.

F. Novelty Detection Test

The novelty detection test utilized novelty detection procedure to signal an update to the base classifier when any new pattern is detected. Any new instances that cannot be classified by the base classifier is marked as '*unknown'* and sent to the novelty detection procedure. The novelty detection procedure executes a *k*-means clustering algorithm to determine whether the '*unknown'* instance is a new label or otherwise. If it is a new label, the base classifier will be updated accordingly.

Similar to the previous proposed method using clustering similarities, the novelty detection test proposed by [14] also employed *k*-means clustering and will suffer the same issues of determining number of clusters, outlier sensitivity, and processing time.

G. Plover Algorithm

In the Plover algorithm proposed by [15], statistical metrics specifically the means, variances, skewness and kurtosis, are used as the measures to support the detection of drifts. The Power Spectrum analysis is also employed as a measure to detect drift in terms of data amplitudes and frequencies.

The proposed algorithm is capable of detecting drifts and estimate its window length; however, it requires selection of appropriate metrics, according to different application domains. For instance, means is known to be sensitive to outliers. Further studies may extend this algorithm to consider other metrics related to statistical learning theory.

H. Chernoff Bound

The Chernoff bound is a probability theory that refers to the total amount of instances that deviates from the mean value. In order to detect concept drift, the authors in [17] utilized the Chernoff bound to define the number of instances that deviates from the mean value in a set of data streams blocks. The key step of this approach is to determine the total amount of instances needed to indicate that the learning algorithm has expired and a new one should be learned from data. This approach is sensitive to the size of data blocks, where small data blocks cause detection error. Future work should investigate the estimation of the number of instances for training data, where new instances could appear or deleted from the data stream.

V. Conclusion

Based on the research questions specified in the earlier section of this paper, a total of 15 articles are categorized in terms of its learning type, algorithms, and drift detection monitoring method. Most of the drift detection methods were applied in a supervised learning setting with various use of drift indication methods. The methods can be divided into two approaches: one using statistical approach, and another approach is based on machine learning. The Kolmogorov–Smirnov test, Jaccard distance, Plover algorithm and Chernoff bound are based on statistical theory, while the clustering similarities, confidence vote, margin density, and novelty detection test use machine learning approach. Out of all the identified concept drift detection methods for unlabeled data streams, the most employed method is the Kolmogorov–Smirnov test, a data-distribution based method capable of monitoring distribution change in unlabeled data streams.

A large-scale comparison of labeled concept drift detection methods has been presented in [25]. Future work may focus on performing experimental evaluation of the methods in this paper in unsupervised learning setting.

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